Where the marks are

Ten systems carry the DOH IM specialist paper. Study them in this order, not in textbook order — the top four are 52% of your score, and all of Dermatology is worth a third of Cardiology.

60%
Pass mark you are aiming past

52%
Sits in the top four systems

90 Q
Roughly what 60% of 150 looks like

Cardiology	ACS pathway & reperfusion windows · AF rate vs rhythm + CHA_2DS_2-VASc · HFrEF four pillars · murmurs · ECG blocks and axis	18%	≈27 Q
Endocrinology	T2DM agent choice by comorbidity · DKA vs HHS · thyroid function patterns · adrenal insufficiency · osteoporosis · Cushing / phaeo screening	12%	≈18 Q
Respiratory Medicine	Asthma vs COPD escalation ladders · CURB-65 and CAP antibiotics · PE probability scoring · pleural effusion (Light's) · O_2 targets	12%	≈18 Q
Nephrology	AKI staging and pre-renal vs ATN · CKD staging and referral · hyperkalaemia sequence · acid–base with anion gap · nephrotic vs nephritic	10%	≈15 Q
Gastroenterology	Upper GI bleed and variceal care · decompensated cirrhosis (SBP, HE, ascites) · IBD flares · coeliac serology · LFT patterns · pancreatitis	10%	≈15 Q
Rheumatology	Acute gout vs septic arthritis · SLE and antibody patterns · giant cell arteritis · vasculitis (ANCA) · seronegative spondyloarthropathy	8%	≈12 Q
Neurology	Stroke thrombolysis window vs thrombectomy · SAH work-up · seizure and status ladder · MS and GBS · headache red flags · neuropathy	8%	≈12 Q
Haematology	Anaemia by MCV and iron studies · transfusion thresholds · VTE and anticoagulation choice · myeloma triad · leukaemia and lymphoma basics	8%	≈12 Q
Infectious Disease	Sepsis bundle timing · TB diagnosis and regimen · HIV screening and prophylaxis · endocarditis (Duke) · meningitis · travel fever	8%	≈12 Q
Dermatology	Drug eruptions and SJS/TEN · cellulitis vs necrotising fasciitis · psoriasis · skin signs of internal disease · pigmented lesion red flags	6%	≈9 Q

READ IT THIS WAY

A system's weight is your hour budget. Two hours on Dermatology is two hours not spent on Cardiology.

THE HONEST CAVEAT

DOH publishes no official weighting. These are DOHPass estimates from content specs and item-frequency analysis — a priority order, not gospel.

CHECK YOUR PAPER

Question counts assume a 150-item paper. Confirm the count on your booking confirmation and rescale.

Six weeks out

Heaviest systems first, while your attention is cheapest. The last week buys nothing new — it only converts what you already half-know into marks.

THE DAILY RHYTHM — SAME EVERY DAY, ALL SIX WEEKS

40 questions
Timed block in the week's system. Timed from day one.

Full review
Every item — including the ones you guessed right.

Leak list
One line per miss: stem clue, your pick, the actual rule.

15 min recall
Flashcards on last week's system, not this week's.

W–6
18% OF PAPER

Baseline & Cardiology

Sit a full diagnostic on day one — before you revise anything. You need a real weakness map, not a felt one.

ACS pathway and reperfusion windows · AF rate vs rhythm, CHA_2DS_2-VASc, DOAC vs warfarin · HFrEF four pillars · murmur-to-lesion mapping · ECG blocks.

W–5
24% OF PAPER

Endocrine & Respiratory

The two biggest blocks after Cardiology. Both are algorithm-heavy, which means they are learnable fast.

T2DM agent choice by comorbidity · DKA vs HHS criteria and fluid order · thyroid patterns · adrenal insufficiency · asthma and COPD ladders · CURB-65 · PE probability.

W–4
20% OF PAPER

Nephrology & Gastroenterology

Where most candidates leak marks on numbers rather than concepts. Learn the cutoffs cold.

AKI staging, pre-renal vs ATN urinary indices · hyperkalaemia sequence · anion-gap acid–base · decompensated cirrhosis (SBP, HE, ascites) · upper GI bleed · IBD flare.

W–3
16% OF PAPER

Rheumatology & Neurology

Pattern-recognition systems. Drill stems, not textbooks — the marks are in the clue-to-diagnosis jump.

Gout vs septic arthritis · SLE and antibody patterns · giant cell arteritis (treat before biopsy) · ANCA vasculitis · stroke windows · status epilepticus ladder · headache red flags.

W–2
22% OF PAPER

Haematology, ID & Dermatology

Three lighter systems together. Derm is only 6% — cap it at two days and move on.

Anaemia by MCV and iron studies · transfusion thresholds · VTE choice and duration · sepsis bundle timing · TB and HIV basics · endocarditis · drug eruptions and SJS/TEN.

W–1
NO NEW TOPICS

Mocks and leaks only

Two full timed mocks, spaced. Everything between them is rebuilt from your own wrong answers.

Days 7 and 4: full mock, full review. Days 6, 5, 3: leak list plus the cram sheets in this pack. Day 2: algorithm sheets out loud. Day before: one light 20-Q block, then stop.

RIGHT ANSWER, WRONG REASON
A guess you got right is a miss that hasn't happened yet. Mark it in the leak list the same as a wrong answer.

NEVER FINISH ON A MISS
Got one wrong at the end of a block? Do five more of that subtopic before you close the laptop. Fewer weeks means shorter blocks, same order.

Cutoffs and thresholds

The values a stem hangs its answer on. If you can't produce the number, you can't rule out the distractor built around it.

CARDIOLOGY

Hypertension, clinic / ABPM daytime	**≥140/90 · ≥135/85**
LDL target, established ASCVD	**<1.4 mmol/L + ≥50% drop**
LDL target, high risk	**<1.8 mmol/L**
STEMI — primary PCI from first contact	**≤90 min**
Lyse instead if PCI unavailable within	**120 min · door-to-needle ≤30**
Reperfusion window from symptom onset	**≤12 h**
HFrEF / HFmrEF / HFpEF by LVEF	**≤40 · 41–49 · ≥50%**
Anticoagulate AF — CHA_2DS_2-VASc	**≥2 men · ≥3 women**

ENDOCRINE

Diabetes — HbA1c / FPG / 2-h OGTT	**≥6.5% · ≥7.0 · ≥11.1**
Prediabetes — HbA1c / fasting	**5.7–6.4% · 5.6–6.9**
DKA triad	**Glucose >13.9 · pH <7.3 · HCO_3 <18 · ketones ≥3.0**
HHS	**Glucose >33.3 · osm >320 · pH >7.3 · ketones low**
Hypoglycaemia — alert / significant	**<3.9 · <3.0 mmol/L**
Treat subclinical hypothyroid when TSH	**>10 mIU/L**
Adrenal insufficiency — 9 am cortisol	**<100 likely · >400 nmol/L excludes**

RESPIRATORY

SpO_2 target — usual / CO_2 retention risk	**94–98% · 88–92%**
CURB-65 — home / admit / ICU review	**0–1 · 2 · ≥3**
Acute severe vs life-threatening asthma	**PEF 33–50% · <33% or SpO_2 <92%**
Type 2 respiratory failure — $PaCO_2$	**>6.0 kPa (45 mmHg)**
NIV in COPD when pH	**<7.35 with $PaCO_2$ >6.0**
Light's — exudate if any one	**Pr ratio >0.5 · LDH ratio >0.6 · LDH >⅔ ULN**
Wells — PE likely	**>4 points**

NEUROLOGY

IV thrombolysis window	**≤4.5 h from onset**
Thrombectomy, anterior LVO	**≤6 h · to 24 h if imaging-selected**
BP ceiling before thrombolysis	**<185/110 mmHg**
Status epilepticus starts at	**5 min of seizure**
LP for xanthochromia — earliest	**12 h after headache onset**

RENAL & ELECTROLYTES

AKI stage 1 — creatinine	**×1.5–1.9 in 7 d or ↑≥26.5 µmol/L in 48 h**
AKI stage 1 — urine output	**<0.5 mL/kg/h for 6–12 h**
AKI stage 2 / stage 3	**×2.0–2.9 · ×3.0 or ≥353.6 or RRT**
Pre-renal vs ATN — FeNa	**<1% · >2%**
CKD G3a / G3b / G4 / G5 — eGFR	**45–59 · 30–44 · 15–29 · <15**
Hyperkalaemia — emergency at	**≥6.5 or any ECG change**
Max Na correction in 24 h	**≤8–10 mmol/L**
Corrected Ca (mmol/L)	**measured + 0.02 × (40 − albumin)**
Normal anion gap	**8–12 mmol/L**

HAEMATOLOGY & VTE

Transfuse — stable / ACS	**Hb <70 · <80 g/L**
Platelets — prophylactic / procedure	**<10 · <50 $\times 10^9$/L**
MCV — micro / normo / macro	**<80 · 80–100 · >100 fL**
Iron deficiency — ferritin	**<30 µg/L (<100 if inflamed)**
Anticoagulation — provoked VTE	**3 months**
INR target — usual / mechanical mitral	**2.5 · 3.0**
Myeloma CRAB	**Ca >2.75 · Cr >177 · Hb <100 · lytic bone**

SEPSIS & INFECTION

qSOFA ≥2 of	**RR ≥22 · altered mentation · SBP ≤100**
Antibiotics from recognition	**within 1 h**
Fluid if hypotensive or lactate ≥4	**30 mL/kg crystalloid**
Septic shock	**Pressors for MAP ≥65 + lactate >2**
Neutropenia / severe — ANC	**<1.0 · <0.5 $\times 10^9$/L**
Neutropenic fever	**≥38.3 °C once or ≥38.0 for 1 h**
SBP — ascitic neutrophils	**≥250 /mm³**

LIVER & GI

Glasgow-Blatchford — consider discharge	**0–1**
Endoscopy in upper GI bleed	**≤24 h · ≤12 h if variceal/unstable**
SAAG — portal hypertension	**≥11 g/L**
AST:ALT suggesting alcohol	**>2:1**
Severe alcoholic hepatitis — Maddrey	**≥32**

Units are SI (mmol/L, µmol/L, g/L, kPa) unless stated. Values are consensus guideline figures for revision — where your local Abu Dhabi or hospital protocol differs, the local protocol wins on the ward and the guideline figure wins on the paper. Re-check anything you plan to rely on against the current guideline edition.

Drug of choice, and what changes it

Most management stems are decided by the third column — the comorbidity, allergy or contraindication that moves the answer off the default.

CARDIOVASCULAR

SITUATION	FIRST LINE	WHAT MOVES THE ANSWER
Hypertension, under 55, not Black African/Caribbean	ACE inhibitor or ARB	Age ≥55 or Black African/Caribbean → CCB first
Hypertension with diabetes or proteinuria	ACE inhibitor or ARB, any age	Pregnancy → stop; labetalol, nifedipine or methyldopa
Heart failure, reduced EF	ARNI (or ACEi) + beta-blocker + MRA + SGLT2i	Loop diuretic treats congestion, not mortality
Acute coronary syndrome	Aspirin 300 mg + P2Y12 + anticoagulant + high-intensity statin	Nitrate and morphine are symptomatic only
AF — stroke prevention	DOAC	Mechanical valve or moderate–severe mitral stenosis → warfarin

ENDOCRINE

SITUATION	FIRST LINE	WHAT MOVES THE ANSWER
Type 2 diabetes, no complications	Metformin + lifestyle	Stop below eGFR 30; hold in acute illness or contrast
Type 2 diabetes with ASCVD, HF or CKD	SGLT2 inhibitor, regardless of HbA1c	Euglycaemic DKA — glucose can be near normal
Type 2 diabetes with obesity	GLP-1 receptor agonist	Avoid with medullary thyroid cancer or MEN2
Diabetic ketoacidosis	IV fluids first, then fixed-rate insulin 0.1 u/kg/h	K^+ below 3.5 → replace before starting insulin
Thyroid storm	Propylthiouracil or carbimazole + beta-blocker + steroid	Iodine only ≥1 h after the thionamide
Adrenal crisis	IV hydrocortisone 100 mg + fluids	Treat first — never wait for the synacthen result

RESPIRATORY & INFECTION

SITUATION	FIRST LINE	WHAT MOVES THE ANSWER
Asthma maintenance	ICS-formoterol as MART	SABA alone is never the correct maintenance answer
COPD maintenance	LABA + LAMA	Add ICS if eosinophils high or exacerbations recur
Community-acquired pneumonia	Amoxicillin if low severity; co-amoxiclav + macrolide if moderate–severe	Penicillin allergy or atypical picture → macrolide
Clostridioides difficile	Oral vancomycin or fidaxomicin	Stop the culprit antibiotic; loperamide is wrong
Tuberculosis, new case	RIPE 2 months, then rifampicin + isoniazid 4 months	Ethambutol → baseline visual acuity; isoniazid → pyridoxine

NEUROLOGY & RHEUMATOLOGY

SITUATION	FIRST LINE	WHAT MOVES THE ANSWER
Status epilepticus	IV lorazepam, repeat once	Then levetiracetam, valproate or phenytoin; then anaesthesia
Acute gout	NSAID, colchicine or steroid	Continue allopurinol if already on it; never start it mid-flare
Giant cell arteritis	High-dose prednisolone immediately	Visual loss → IV methylprednisolone; biopsy after starting
SLE	Hydroxychloroquine for everyone	Organ involvement → steroid plus immunosuppressant

EMERGENCIES & ANTIDOTES

SITUATION	FIRST LINE	WHAT MOVES THE ANSWER
Anaphylaxis	IM adrenaline 0.5 mg of 1:1000, anterolateral thigh	Repeat at 5 min; steroid and antihistamine are adjuncts
Hyperkalaemia with ECG changes	IV calcium gluconate	Calcium protects the myocardium — it does not lower K^+
Paracetamol overdose	N-acetylcysteine	Staggered or unknown timing → treat without the nomogram
Warfarin with major bleeding	IV vitamin K + prothrombin complex concentrate	FFP only when PCC is unavailable
Beta-blocker or calcium-channel blocker overdose	Glucagon, or high-dose insulin euglycaemic therapy	Add IV calcium for the calcium-channel blocker

Antibiotic choices vary by local resistance and formulary — for the exam answer the guideline default, on the ward answer your hospital protocol. Doses shown are adult unless stated.

The clue, the call, the next move

Written examiners plant one phrase that settles the diagnosis. Learn the phrase and the immediate next step — the option list is usually four plausible investigations and one right order.

CARDIOVASCULAR & RESPIRATORY

PHRASE IN THE STEM	THINK	NEXT MOVE
Tearing chest pain radiating to the back, blood pressure different in each arm	**Aortic dissection**	CT aortogram. Control heart rate with IV labetalol before any vasodilator.
Pleuritic pain and sudden breathlessness after immobility, surgery or a long flight	**Pulmonary embolism**	Wells score, then CTPA. Anticoagulate while waiting if probability is high.
Raised JVP, muffled heart sounds, hypotension, pulsus paradoxus	**Cardiac tamponade**	Urgent bedside echo, then pericardiocentesis.
Syncope on exertion with an ejection systolic murmur radiating to the carotids	**Severe aortic stenosis**	Echo. Avoid nitrates and other vasodilators.
COPD patient becomes drowsy after starting high-flow oxygen	**Hypercapnic respiratory failure**	ABG. Titrate O_2 to 88–92%; NIV if pH is below 7.35.

NEUROLOGY

PHRASE IN THE STEM	THINK	NEXT MOVE
Thunderclap headache, worst of their life, CT head reported normal	**Subarachnoid haemorrhage**	Lumbar puncture at 12 h or later for xanthochromia.
Sudden loss of vision in one eye, jaw claudication, scalp tenderness, high ESR	**Giant cell arteritis**	High-dose steroid immediately — biopsy afterwards, never before.
Back pain with leg weakness or bladder change in a patient with known cancer	**Metastatic cord compression**	Whole-spine MRI urgently plus dexamethasone. Hours matter.
Ascending weakness with absent reflexes days after a diarrhoeal illness	**Guillain–Barré syndrome**	Serial FVC — not oxygen saturation. IVIG or plasma exchange.
Headache worse lying flat, morning vomiting, papilloedema	**Raised intracranial pressure**	Imaging before lumbar puncture, always.

ENDOCRINE & METABOLIC

PHRASE IN THE STEM	THINK	NEXT MOVE
Euvolaemic hyponatraemia, urine osmolality above 100, urine sodium above 30	**SIADH**	Fluid restriction, then hunt the cause: drug, lung, CNS or malignancy.
Fatigue, skin pigmentation, low sodium, high potassium, postural drop	**Primary adrenal insufficiency**	9 am cortisol and short synacthen — but give hydrocortisone first if unwell.
Episodic headache, sweating and palpitations with labile blood pressure	**Phaeochromocytoma**	Plasma or 24 h urinary metanephrines. Alpha-blockade before beta-blockade.
Acidotic and unwell on an SGLT2 inhibitor with near-normal glucose	**Euglycaemic DKA**	Check ketones and ABG. Treat as DKA — the glucose reassures nobody.
Fever, agitation and fast AF in a patient with Graves after an infection	**Thyroid storm**	Thionamide, then beta-blocker and steroid. Iodine only 1 h after the thionamide.

GASTRO, INFECTION, RHEUMATOLOGY, SKIN

PHRASE IN THE STEM	THINK	NEXT MOVE
Cirrhosis with ascites who becomes febrile, confused or tender	**Spontaneous bacterial peritonitis**	Diagnostic tap. Neutrophils $\geq 250/mm^3$ → antibiotics plus albumin.
Profuse watery diarrhoea days into a course of antibiotics	**Clostridioides difficile**	Stool toxin, oral vancomycin, stop the culprit. Never an antimotility drug.
One hot, swollen, exquisitely painful joint with fever	**Septic arthritis until proven otherwise**	Aspirate the joint before the first dose of antibiotic.
Target lesions with mouth and eye ulceration after starting a new drug	**Stevens–Johnson / TEN**	Stop the drug immediately; burns-unit level supportive care.
Haemoptysis with haematuria and a rising creatinine	**Pulmonary–renal syndrome**	ANCA and anti-GBM, urgent renal referral. Do not wait for the biopsy.

Oncology triggers sit on sheet 9 with their emergencies. When two options are both correct investigations, the answer is the one that changes management soonest.

Chest pain, fast heart, breathless

Management-order questions live here. The mark is for the step that comes *next* — not the step that is most thorough.

Acute coronary syndrome

Ischaemic pain with diaphoresis, nausea or radiation. The ECG and the troponin timing decide everything else.

1. **ECG within 10 minutes of arrival**
   Repeat if pain continues and the first is non-diagnostic. ST depression in V1–V3 → posterior leads before calling it NSTEMI.
2. **Aspirin 300 mg, a P2Y12 inhibitor and an anticoagulant**
   Nitrate and morphine relieve symptoms only. Oxygen just for hypoxia.
3. **STEMI → primary PCI within 90 minutes of first medical contact**
   Not achievable inside 120 minutes → fibrinolyse with a door-to-needle time of 30 minutes or less, then transfer.
4. **NSTEMI or unstable angina → risk-stratify, then time the angiogram**
   High GRACE risk or ongoing ischaemia → angiography within 24 hours. Everyone leaves on a high-intensity statin, DAPT, ACE inhibitor, beta-blocker and an echo.

TRAP A single normal troponin never excludes ACS. When every option is a sensible test, the answer is usually the repeat troponin at the right interval.

Atrial fibrillation with a fast ventricular rate

Irregularly irregular pulse above 110. The first question is haemodynamic, never pharmacological.

1. **Unstable? Synchronised DC cardioversion now**
   Shock, syncope, myocardial ischaemia or acute heart failure — do not rate-control your way through it.
2. **Stable → rate control with a beta-blocker or diltiazem**
   Digoxin suits the sedentary patient or heart failure; amiodarone if rhythm control is intended.
3. **Time the onset before considering cardioversion**
   Clearly under 48 hours → cardioversion is on the table. 48 hours or unknown → three weeks of anticoagulation, or a TOE first.
4. **Score the stroke risk separately, then find the driver**
   CHA_2DS_2-VASc ≥2 men or ≥3 women → DOAC; mechanical valve or moderate–severe mitral stenosis → warfarin. Look for sepsis, thyrotoxicosis, PE or alcohol.

TRAP Never pair verapamil or diltiazem with a beta-blocker, and avoid both in decompensated heart failure with reduced ejection fraction.

Suspected pulmonary embolism

Pleuritic pain, breathlessness, tachycardia — usually with immobility, surgery, malignancy or pregnancy in the stem.

1. **Score the probability first**
   Wells: PE unlikely → D-dimer. PE likely → straight to CTPA, no D-dimer.
2. **Unstable → bedside echo for right ventricular strain**
   Peri-arrest or persistent hypotension with a convincing story → thrombolyse without waiting for the CTPA.
3. **Anticoagulate while you wait when probability is high**
   Unless active bleeding or a clear contraindication makes the risk unacceptable.
4. **Confirmed → DOAC, then set the duration before discharge**
   Severe renal impairment or antiphospholipid syndrome → LMWH into warfarin. Provoked 3 months; unprovoked or active cancer at least 3, extended therapy considered.

TRAP A negative D-dimer only helps a low-probability patient. In pregnancy the pathway is different — chest X-ray and leg dopplers come first.

Sugar, potassium, sodium

Three emergencies where the sequence itself is the examined content, and where correcting too fast is its own wrong answer.

Diabetic ketoacidosis

Glucose >13.9 (or near-normal on an SGLT2 inhibitor), pH <7.3, bicarbonate <18, ketones ≥3.0. HHS is the mirror image — glucose >33.3, osmolality >320, few ketones, fluids first and slower.

1. **Fluid before insulin — 0.9% sodium chloride**
   The deficit is often 5–7 litres. Restore the circulation first; the insulin can wait the few minutes it takes.
2. **Fixed-rate insulin at 0.1 units/kg/h**
   Keep the patient's usual long-acting basal insulin running alongside it. Do not stop it.
3. **Potassium sets the pace**
   Below 3.5 → hold the insulin and replace first. Between 3.5 and 5.5 → 40 mmol/L in the bag. Above 5.5 → none, and recheck.
4. **Add 10% glucose below 14, and treat to ketone clearance**
   Resolution is ketones <0.6, pH >7.3, bicarbonate >15. Convert to subcutaneous only when eating, with an overlap.

TRAP The endpoint is ketone clearance, not the glucose. Bicarbonate is almost never the right option, and over-fast correction risks cerebral oedema.

Hyperkalaemia

Potassium above 6.0, or any level with ECG change — peaked T waves, flat or absent P waves, a widening QRS.

1. **ECG first, immediately**
   Repeat the sample only if it costs no time. A haemolysed specimen is a real answer — but never when the ECG is abnormal.
2. **ECG change or K ≥6.5 → IV calcium gluconate now**
   10 mL of 10%, repeated if changes persist. It stabilises the myocardium; it does not lower the potassium at all.
3. **Shift it inside the cells**
   10 units of soluble insulin in 25 g of glucose, plus nebulised salbutamol. Then watch the glucose for several hours.
4. **Remove it, and stop what caused it**
   ACE inhibitor, ARB, spironolactone, NSAID, trimethoprim. Potassium binder for the sub-acute problem; dialysis if refractory or in established renal failure.

TRAP Calcium buys minutes, insulin-dextrose buys hours, only removal is definitive. A question that stops at insulin-dextrose has not finished treating the patient.

Hyponatraemia

Sodium below 135. Volume status and paired osmolalities decide the answer — the number alone decides nothing.

1. **Severe symptoms → hypertonic 3% saline, whatever the cause**
   Seizure, marked drowsiness or a falling GCS. Give it as boluses and stop chasing the aetiology until the brain is safe.
2. **Otherwise assess volume status and send the right tests**
   Paired serum and urine osmolality with a urine sodium, plus thyroid and cortisol if the picture fits.
3. **Treat by compartment**
   Hypovolaemic with urine Na <20 → 0.9% saline. Euvolaemic, urine osm >100 and urine Na >30 → SIADH: restrict fluid, find the drug, lung, CNS or malignant cause. Hypervolaemic → treat the failing organ.
4. **Cap the correction at 8–10 mmol/L in 24 hours**
   Lower still if malnourished, alcohol-dependent, hypokalaemic, or the sodium has been low for a long time.

TRAP Over-rapid correction causes osmotic demyelination. Exclude pseudohyponatraemia — high glucose, paraprotein or lipids — before treating anything.

Sepsis, bleeding, failing kidneys

The three acute-take pathways where a timing figure — one hour, twenty-four hours, forty-eight — is usually the marked point.

Sepsis

Infection plus organ dysfunction. qSOFA ≥2 — respiratory rate ≥22, altered mentation, systolic ≤100 — is a prompt to act, not a diagnosis.

1. **Recognise it and start the clock**
   Senior review early. Do not wait for a lactate result before beginning treatment.
2. **Cultures, lactate, urine output — then oxygen, antibiotics, fluids**
   Blood cultures before the first dose, but only if that costs no delay. Broad-spectrum IV antibiotics within one hour.
3. **30 mL/kg crystalloid if hypotensive or lactate ≥4, then reassess**
   Reassess perfusion after each bolus rather than running fluid blindly.
4. **MAP still under 65 → noradrenaline and critical care, and get source control**
   Drain the collection, remove the line, take them to theatre. Source control is what actually ends the sepsis.

TRAP Antibiotics within the hour beat a perfect microbiological work-up. Never delay the first dose for imaging or a second set of cultures.

Upper gastrointestinal bleed

Haematemesis, coffee-ground vomit or melaena. Chronic liver disease anywhere in the stem changes the whole answer.

1. **Resuscitate and score**
   Two large-bore cannulae, crossmatch, Glasgow-Blatchford. A score of 0–1 in a well patient can be managed without admission.
2. **Transfuse restrictively — target Hb 70 g/L, or 80 with cardiac disease**
   Over-transfusion raises portal pressure and makes variceal bleeding worse.
3. **Suspected varices → terlipressin plus prophylactic antibiotics before endoscopy**
   For non-variceal bleeding, a PPI before endoscopy adds nothing — it belongs after the endoscopic diagnosis.
4. **Endoscopy within 24 hours, within 12 if unstable or variceal**
   Correct coagulopathy, stop the anticoagulant, and reverse it when the bleeding is life-threatening.

TRAP Antibiotics in variceal bleeding are a mortality intervention, not a courtesy. A urea rise out of proportion to the creatinine is the classic upper GI clue.

Acute kidney injury

Creatinine ×1.5 within 7 days, a rise of ≥26.5 μmol/L within 48 hours, or urine output under 0.5 mL/kg/h for 6 hours.

1. **Exclude obstruction before anything clever**
   Catheter and bladder scan; ultrasound within 24 hours when no cause is obvious.
2. **Assess volume and perfusion honestly**
   Hypovolaemic → fluid challenge and reassess. Already overloaded → the answer is not more fluid.
3. **Stop the nephrotoxins and re-dose everything else**
   NSAIDs, ACE inhibitors and ARBs, aminoglycosides, contrast. Review every drug for renal adjustment.
4. **Dip the urine — it is the cheapest test on the sheet**
   Blood and protein without an obvious explanation → glomerulonephritis or vasculitis. Refer the same day.

DIALYSE FOR refractory hyperkalaemia, refractory acidosis, refractory fluid overload, uraemic pericarditis or encephalopathy, or a dialysable poison.

Oncological emergencies

Each clue line below is the phrase the stem will use. These five carry more marks than their share of the blueprint suggests, because every one of them is a timed decision.

Neutropenic sepsis

Fever ≥38.3 °C once, or ≥38.0 for an hour, within six weeks of chemotherapy.

1. **Antibiotics within one hour**
   Piperacillin-tazobactam as the usual first line. Do not wait for the neutrophil count.
2. **Culture peripherally and from every line lumen**
   Full septic screen, then the rest of the sepsis bundle as normal.
3. **Escalate or step down deliberately**
   Vancomycin for line infection or MRSA risk. Score with MASCC — low-risk patients can move to oral therapy.

TRAP A normal white cell count on arrival excludes nothing. The first dose goes in before the FBC comes back.

Tumour lysis syndrome

Bulky chemo-sensitive tumour, days into treatment: high potassium and phosphate, high urate, low calcium, rising creatinine.

1. **Stratify risk before cycle one**
   High-grade lymphoma, ALL, high white count, bulky disease, existing renal impairment.
2. **Fluids for everyone at risk, then urate control**
   Rasburicase for high risk, allopurinol for lower risk.
3. **Treat the potassium first**
   It kills before the urate does. Cardiac monitoring, and renal input early for refractory disturbance.

TRAP Rasburicase is contraindicated in G6PD deficiency, and calcium for asymptomatic hypocalcaemia risks precipitation with phosphate.

Hypercalcaemia of malignancy

Known cancer with confusion, constipation, polyuria and thirst, or a corrected calcium above 3.0 mmol/L.

1. **Correct for albumin before you react**
   Measured calcium + 0.02 × (40 − albumin in g/L).
2. **IV 0.9% saline is the treatment, not the preparation**
   Rehydration alone brings the calcium down substantially.
3. **Then a renally-adjusted IV bisphosphonate**
   Zoledronic acid; the nadir takes 2–4 days. Denosumab if refractory or in renal failure.

TRAP Loop diuretics are not routine — only once fluid-replete and overloaded. Stop thiazides, calcium and vitamin D.

Metastatic cord compression

New or progressive back pain with leg weakness, a sensory level, or bladder and bowel change in a patient with cancer.

1. **Dexamethasone immediately**
   Typically 16 mg daily with gastric protection — before the imaging is even booked.
2. **Whole-spine MRI within 24 hours**
   Sooner if the neurology is progressing. Whole spine, because multiple levels are common.
3. **Same-day radiotherapy or surgical decision**
   Nurse flat until the spine is declared stable.

TRAP Function at the moment of treatment predicts function afterwards. Once they cannot walk, they usually will not walk again.

Superior vena cava obstruction

Facial and arm swelling with distended neck and chest-wall veins, breathlessness worse on stooping — usually with a known or suspected intrathoracic primary.

1. **Assess the airway first — stridor is what makes this a true emergency**
   Sit the patient up, give oxygen, and start dexamethasone.
2. **CT chest with contrast to define the level, the cause and any thrombus**
   Get a tissue diagnosis where the patient can wait for it — the histology dictates the definitive treatment.
3. **Endovascular stenting for rapid relief, then treat the tumour**
   Radiotherapy or chemotherapy as the histology directs.

TRAP In a chemo-sensitive tumour — small-cell, lymphoma, germ cell — systemic treatment relieves it quickly, so an undiagnosed patient needs a biopsy, not just a steroid.

Anaemia, haemolysis, the film

Haematology questions are pattern questions. Nearly all of them resolve to one table, one index, or one thing you can see down a microscope.

MCV IS THE FIRST FORK

Microcytic <80 fL
Iron deficiency · thalassaemia · chronic disease (late) · sideroblastic · lead

Normocytic 80–100
Acute blood loss · chronic disease · renal · haemolysis · marrow failure · mixed deficiency

Macrocytic >100
Megaloblastic: B12, folate. Non-megaloblastic: alcohol, liver, hypothyroid, myelodysplasia, reticulocytosis, drugs

IRON STUDIES — THE TABLE THEY BUILD DISTRACTORS FROM

PICTURE	FERRITIN	SERUM IRON	TIBC	TRANSFERRIN SAT
Iron deficiency	↓	↓	↑	↓ <20%
Anaemia of chronic disease	↑ / normal	↓	↓	normal / ↓
Thalassaemia trait	normal	normal	normal	normal
Haemochromatosis	↑↑	↑	↓	↑↑ >45%
Sideroblastic	↑	↑	normal	↑

GEMS Ferritin is an acute-phase reactant — with inflammation the threshold moves from 30 to 100 µg/L, so a "normal" ferritin excludes nothing. Mentzer index (MCV ÷ RBC): under 13 is thalassaemia trait, over 13 is iron deficiency — thal trait runs a *high* red cell count with a normal RDW. New iron deficiency in a man or a postmenopausal woman means endoscopy at both ends.

B12 AND FOLATE — WHERE THE MARKS ACTUALLY ARE

Neurology — subacute combined degeneration of the cord	**B12 only, never folate**
Which to replace first	**B12 before folate, always**
Homocysteine / methylmalonic acid	**Both raise homocysteine · only B12 raises MMA**
Pernicious anaemia antibodies	**Anti-intrinsic-factor specific · anti-parietal-cell sensitive**
Culprits to look for in the stem	**Metformin · PPI · ileal Crohn's or resection · vegan diet · nitrous oxide**
Film in either deficiency	**Hypersegmented neutrophils, >5 lobes**

TRAP Giving folate to someone B12-deficient can precipitate or worsen the cord lesion. If a stem offers both, the order is the question.

HAEMOLYSIS — SCREEN, THEN SPLIT ON THE DAT

The screen, all four together: **↑ reticulocytes · ↑ LDH · ↑ unconjugated bilirubin · ↓ haptoglobin**

TYPE	CLUE	WHAT FOLLOWS
Warm autoimmune — IgG, extravascular	**DAT positive**	Look for CLL, lymphoma, SLE or methyldopa. Steroids first line.
Cold agglutinin — IgM, complement	**DAT positive**	Mycoplasma, EBV, Waldenström. Keep warm, rituximab — steroids disappoint.
G6PD deficiency — X-linked	**DAT negative**	Oxidant drugs, fava beans, infection. Bite cells and Heinz bodies.
Hereditary spherocytosis	**DAT negative**	Raised MCHC, EMA binding test. Spherocytes without a positive DAT.
Microangiopathic (MAHA)	**DAT negative**	Schistocytes → TTP, HUS, DIC, HELLP, malignant hypertension, mechanical valve.

GEMS The G6PD level is falsely normal *during* the acute episode — the deficient cells have already lysed, so retest at three months. In sickle cell, acute chest syndrome is the leading cause of death: fever, chest pain and a new infiltrate mean antibiotics, oxygen, analgesia and transfusion, not observation.

THE FILM — ONE LOOK, ONE ANSWER

Schistocytes	**MAHA — TTP, HUS, DIC**
Spherocytes	**Spherocytosis or warm AIHA**
Bite cells, Heinz bodies	**G6PD deficiency**
Target cells	**Thalassaemia, liver disease, post-splenectomy**
Howell-Jolly bodies	**Hyposplenism**
Basophilic stippling	**Lead, thalassaemia, sideroblastic**
Tear-drop cells, leucoerythroblastic	**Myelofibrosis or marrow infiltration**
Rouleaux	**Myeloma, very high ESR**
Auer rods	**Acute myeloid leukaemia**
Smudge cells	**Chronic lymphocytic leukaemia**

Platelets, clotting, transfusion

Three tables carry most of this topic — and three of the traps below reverse the intuitive answer, which is exactly why they get written.

READING THE COAGULATION SCREEN

PICTURE	PT / INR	APTT	PLATELETS	WHAT SETTLES IT
Haemophilia A or B	normal	↑	normal	X-linked, bleeding into joints and muscles. Factor VIII or IX assay.
von Willebrand disease	normal	↑ / normal	normal	Commonest inherited bleeding disorder. Mucosal and menstrual bleeding.
Warfarin	↑	normal / ↑	normal	Factors II, VII, IX and X. PT moves first because VII is shortest-lived.
Unfractionated heparin	normal / ↑	↑	normal	Falling platelets on heparin is HIT until disproved, not dilution.
DIC	↑	↑	↓	Fibrinogen low, D-dimer high, and **factor VIII low**.
Liver disease	↑	↑	↓	Looks identical — except **factor VIII is normal or high**. That is the discriminator.
Vitamin K deficiency	↑	normal → ↑	normal	Corrects with vitamin K. Factor VIII normal, fibrinogen normal.
Immune thrombocytopenia	normal	normal	↓↓	Isolated. Anything else abnormal means it is not ITP.

GEMS A mixing study that *corrects* means factor deficiency; one that *fails* to correct means an inhibitor. An isolated long APTT with **THROMBOSIS RATHER THAN BLEEDING** is antiphospholipid syndrome — the lupus anticoagulant prolongs the test in the tube while clotting the patient. With no bleeding history at all, it is factor XII deficiency, clinically silent.

LOW PLATELETS — FOUR DIAGNOSES, FOUR DIFFERENT MOVES

DIAGNOSIS	THE GIVEAWAY	WHAT TO DO
Immune thrombocytopenia (ITP)	**Isolated low platelets, normal film**	Treat only below 30 ×10^9/L or if bleeding: steroid ± IVIG. IVIG raises the count faster.
Thrombotic thrombocytopenic purpura	**Schistocytes with low platelets**	ADAMTS13 under 10%. Urgent plasma exchange — the pentad is a description, not an entry requirement.
Heparin-induced thrombocytopenia	**Platelets fall >50% on day 5–10**	Score with 4Ts. Stop every heparin including line flushes, start a non-heparin anticoagulant.
Disseminated intravascular coagulation	**Everything deranged at once**	Treat the precipitant — sepsis, obstetric, malignancy. Products only if actively bleeding.

THE TWO THAT REVERSE In TTP, platelet transfusion is **CONTRAINDICATED** unless bleeding is life-threatening — it feeds the thrombosis. In HIT the patient **CLOTS, NOT BLEEDS**, and warfarin waits until the platelet count recovers, or you risk venous limb gangrene.

TRANSFUSION REACTIONS — TIMING PLUS ONE SIGN

REACTION	ONSET	THE SIGN THAT NAMES IT	ACTION
Acute haemolytic (ABO)	Minutes	Fever, loin pain, dark urine, hypotension	Stop. Saline, recheck the label, tell the lab.
Febrile non-haemolytic	30–60 min	Fever and rigors, otherwise well	Slow or pause, paracetamol.
Allergic, urticarial	Minutes	Itch and urticaria alone	Antihistamine, restart slowly.
Anaphylaxis	Minutes	Shock and wheeze — think IgA deficiency	IM adrenaline. IgA-deficient components in future.
TACO — overload	Within 6 h	Hypertension, **raised JVP**, pulmonary oedema	Stop, sit up, diuretic.
TRALI — lung injury	Within 6 h	Hypoxia and infiltrates with a **normal JVP**	Stop, oxygen, supportive care. No diuretic.
Bacterial contamination	Rapid	High fever and shock — usually platelets	Stop, cultures, broad-spectrum antibiotics.
Delayed haemolytic	5–10 days	Hb falls again, jaundice, positive DAT	Supportive; identify the antibody.

GEM TACO and TRALI share a window and a chest X-ray. Blood pressure and JVP separate them — both up in overload, normal or down in lung injury. A diuretic fixes one and harms the other, which is why it is asked so often.

Leukaemia, lymphoma, myeloma

Each of these carries one signature the examiner can put in a single clause. Learn the clause and the complication it drags behind it.

THE FOUR LEUKAEMIAS

DISEASE	SIGNATURE	WHAT IT DRAGS WITH IT
Acute myeloid (AML)	**Auer rods, blasts ≥20%**	The promyelocytic subtype, t(15;17), arrives in DIC — give ATRA immediately, before the workup is finished.
Acute lymphoblastic (ALL)	**Children, lymphoblasts**	CNS and testis are sanctuary sites, so prophylaxis is part of the regimen. Philadelphia-positive is the poor-prognosis adult form.
Chronic myeloid (CML)	**Philadelphia t(9;22), BCR-ABL**	Massive splenomegaly, the whole myeloid series on the film, low leucocyte alkaline phosphatase. Imatinib; watch for blast crisis.
Chronic lymphocytic (CLL)	**Smudge cells, incidental lymphocytosis**	Warm AIHA and ITP, hypogammaglobulinaemia with recurrent infection, and Richter's transformation to high-grade lymphoma.

HODGKIN AGAINST NON-HODGKIN

	HODGKIN LYMPHOMA	NON-HODGKIN LYMPHOMA
Cell	Reed-Sternberg cells	Diffuse large B-cell is the commonest of many
Spread	Contiguous, node to adjacent node	Non-contiguous, frequently extranodal
Age	Bimodal — young adults and the elderly	Rises steadily with age
Treatment	ABVD — bleomycin means lung toxicity	R-CHOP — doxorubicin means cardiotoxicity

GEMS Nodal pain brought on by alcohol is Hodgkin, and appears in stems precisely because it is unmistakable. B symptoms have an exact definition: fever above 38 °C, drenching night sweats, over 10% weight loss in six months. A bulky lymphoma starting chemotherapy is the classic tumour lysis setup — sheet 9.

MYELOMA AND THE PARAPROTEINS

The diagnostic quartet — CRAB	**Calcium >2.75 · Creatinine >177 · Hb <100 · lytic bone lesions**
On the film and in the blood	**Rouleaux · very high ESR · paraprotein on electrophoresis**
In the urine	**Bence-Jones protein — free light chains**
Why the kidney fails	**Light-chain cast nephropathy — hydrate, stop nephrotoxics**
MGUS, and what it becomes	**Paraprotein <30 g/L, marrow plasma cells <10%, no CRAB · 1% per year**

TWO TRAPS The isotope bone scan is **NEGATIVE** in myeloma — the lesions are purely lytic and provoke no osteoblastic reaction, so imaging is a skeletal survey, whole-body low-dose CT or MRI. And a normal ESR does not exclude it: light-chain-only disease produces no measurable paraprotein band, so send serum free light chains.

MYELOPROLIFERATIVE NEOPLASMS

DISEASE	MARKER	THE CLINICAL CLUE
Polycythaemia vera	**JAK2 V617F**	Itch after a hot bath, plethora, thrombosis. Venesection plus aspirin, hydroxycarbamide if high risk.
Essential thrombocythaemia	**JAK2 or CALR**	Thrombosis — but bleeding at extreme counts, from an acquired von Willebrand defect.
Myelofibrosis	**JAK2, dry tap**	Tear-drop cells and a leucoerythroblastic film with massive splenomegaly.

GEM Erythropoietin separates the polycythaemias: **LOW** EPO means polycythaemia vera, while a high or normal level points to a secondary cause — hypoxia, smoking, renal or hepatic tumour. And of this group only CML is BCR-ABL positive; the other three are defined by JAK2 and its company.

ANTICOAGULATION GEMS THAT SIT OUTSIDE THE DRUG SHEET

Monitoring	**UFH by APTT · LMWH by anti-Xa only in renal impairment, pregnancy or extreme weight**
Reversing heparin	**Protamine reverses UFH fully, LMWH only partially**
Reversing a DOAC	**Idarucizumab for dabigatran · andexanet alfa for the anti-Xa agents**
Massive haemorrhage	**Red cells, plasma and platelets 1:1:1 · tranexamic acid early in trauma**
Why warfarin starts procoagulant	**Protein C falls first — hence bridging, and skin necrosis**